

Ahmed Rafat

Software Engineer (AI/ML)

+601135844762

www.linkedin.com/in/ahmedrafatk

✉ ahmedrafatk@gmail.com

🔗 <https://github.com/sktalt>

PROFESSIONAL SUMMARY

Software Engineer specializing in AI/ML systems, backend engineering, and scalable infrastructure. Experienced building production-grade systems with measurable performance impact, including reducing execution time by 80% through multi-layer parallelization and designing memory-efficient large-scale data processing pipelines. Strong background in system design, algorithm optimization, and applied machine learning research, with published AI research and hands-on experience across backend services, automation, full-stack development, and real-time detection systems. Recognized for rapid technical growth, ownership of high-impact projects, and leading engineering initiatives early in career.

WORK EXPERIENCE

Software Engineer

February 2025 - Present

Juris Technologies Sdn Bhd (Juristech)

Malaysia

- Engineered multi-layer parallel processing optimizations that reduced average execution time from 10 minutes to 2 minutes, improving system throughput by 80% in production workflows
- Designed and implemented secure internal and external APIs enabling cross-product service integration while centralizing maintenance and access control
- Built memory-efficient algorithms for large-scale data processing without increasing PHP memory consumption in production environments
- Led technical execution and deployment coordination for 5 concurrent client implementations
- Developed 6 standalone production components and contributed to more than 20 system modules across backend infrastructure and business logic layers
- Mentored and assessed incoming engineers during onboarding and technical evaluation processes
- Advanced into technical leadership responsibilities within first year through ownership of delivery, architecture discussions, and cross-team coordination

Software Engineer Intern

October 2023 - January 2024

Juris Technologies Sdn Bhd (Juristech)

Malaysia

- Developed automated test coverage for multiple system components, improving validation reliability and reducing manual testing effort
- Built backend utilities and reusable engineering tools that improved internal system performance and maintainability
- Investigated and resolved recurring production-impacting issues affecting user experience and system stability

Software Engineer Intern

September 2022 - December 2022

Juris Technologies Sdn Bhd (Juristech)

Malaysia

- Implemented backend security-focused system components to strengthen platform reliability and protection mechanisms
- Developed backend modules optimized for improved response time and execution efficiency
- Automated repetitive database operations through custom scripting workflows

PERSONAL PROJECTS

Final Year Project - AI malware detection

Jan 2024 - Aug 2024

AI/ML • Cybersecurity • Real-Time Detection • Research

- Built an AI-powered malware detection system for PE files using machine learning classification models
- Achieved peak model accuracy of 99.5% across malware detection experiments
- Designed a real-time TCP packet inspection and malicious traffic detection pipeline with detection accuracy reaching 99.6%

Ahmed Rafat

+601135844762 | ahmedrafatk@gmail.com | <https://github.com/sktalt>

- Developed end-to-end detection workflows including feature extraction, model evaluation, packet analysis, and system integration
- Published associated research work in applied AI and cybersecurity research: [LINK TO THE PAPER](#)

AI PDF RAG Analyzer

December 2025

LLMs • Retrieval-Augmented Generation • System Design

- Developed a Retrieval-Augmented Generation (RAG) system enabling users to query multiple PDF documents using LLM-based semantic retrieval
- Designed document ingestion, chunking, retrieval, and response generation workflows for contextual question answering
- Focused on scalable architecture and extensible AI pipeline design for document intelligence applications

SEMQ platform

September 2024 – March 2025

Full-Stack Engineering • Product Development

- Built a full-stack platform for exchanging academic study materials between students
- Designed and developed frontend, backend, and database architecture independently
- Managed end-to-end product implementation from planning through deployment

Automated IoT fish feeder

May 2023 – April 2024

IoT • Embedded Systems • APIs

- Developed an IoT-based automated fish feeding system with environmental monitoring and mobile application integration
- Contributed to hardware assembly, embedded programming, and backend API development
- Implemented automated tracking and visualization of water-condition metrics

Mixed reality robotic arm

May 2023 – September 2023

AR/VR • Robotics • Interactive Systems

- Contributed to development of a low-cost AR/VR-assisted robotics teaching platform for industrial robotics education
- Supported planning, implementation, and software development for mixed reality integration
- Project received Gold Medal recognition at the IICE 2023 competition hosted by Universiti Teknologi Malaysia

PUBLICATIONS

Published research papers

- Research publication on AI and Big Data integration, [LINK TO PAPER](#)
- Research publication on AI-driven credit card fraud detection, [LINK TO PAPER](#)
- Research publication associated with AI malware detection and network traffic analysis, [LINK TO THE PAPER](#)

TECHNICAL SKILLS

Languages | Python, JavaScript, C/C++, SQL, PHP

AI/ML | TensorFlow, Pandas, NumPy, Machine Learning, Data Analysis, Model Evaluation, RAG Systems

Developer Tools | Git, Docker, Composer, Visual Studio Code

Backend & Infrastructure | API Development, System Design, Parallel Processing, Algorithm Optimization, Database Automation, Docker, Linux

Ahmed Rafat

+601135844762 | ahmedrafatk@gmail.com | <https://github.com/sktalt>

LEADERSHIP & ACTIVITIES

Organizing Chairperson -- Techstars startup weekend

Techstars | Jan 2023 – June 2023

- Led coordination and execution of a 54-hour startup event hosted at UCSI University
- Managed event planning, speaker coordination, and participant competition structure

Event planner -- TheSpark forum

TheSpark | Feb 2024

- Planned the event flow, ensured the marketing is well implemented
- Selected the speaker with relative background to the forum topic

Volunteer -- KL startup summit

KL startup summit | July 2024

- Volunteered to help as crew on the event day

EDUCATION

Bachelor of computer engineering (Artificial Intelligence) with Honours

UCSI university, CGPA 3.73/4.00 (first class with Honours)

Scholarship awardee throughout the full university study duration, 7-time dean list awardee

2024

Malaysia

IELTS English

Band 6.0

2019

Egypt

Bashaer Aljazira High School (KSA)

High school, score 99.97%

2020

KSA

Qiyas test (Saudi Arabia)

Score 96%

2020

KSA